PHYS 8602

Problem #3 Due 3/12/25

Consider the identical system that you used in Homework Problem #2 and carry out a constant temperature MD simulation at the temperature that **you** found as the mean temperature from Problem #2. Show a plot of the time dependence of the kinetic energy, potential energy, and total energy. Once the simulation has reached equilibrium, determine:

a.) the average Kinetic Energy (in reduced units)

b.) the average Potential Energy (in reduced units)

c.) the average total Energy (in reduced units).

How does the average total energy compare with the energy in Problem #2?

Please make your report concise but describe how you analyzed your results and how you reached your conclusions. Make any comments that you feel are relevant. Include an annotated copy of your program and a copy of a summary.log.

Your report should be submitted as a single pdf file to dlandau@uga.edu. The filename should be "your_last name"_PHYS8602_prob3.pdf. Please remember to put your name on your report!